Build RAG Applications: Get Started

Today's Skill Points 90 XP
See skill progress

Module 1
Introduction to RAG

Welcome to the Course

- Course Introduction
  Video • 2 min
- Course Overview
  Reading • 4 min
- RAG and Agentic AI Professional Certificate Overview
  Video • 5 min
- Reading: Helpful Tips for Course Completion
  Ungraded Plugin • 5 min

What is RAG?

- Why RAG?
  Video • 6 min
- More RAG Details
  Video • 7 min
- What is RAG?
  Reading • 15 min
- Understanding the Why and How of RAG
  Dialogue • 15 min
- Summarize Private Documents Using RAG, LangChain, and LLMs
  Ungraded App Item • 45 min
- Building the Case: Explaining RAG Architecture to Your Product Stakeholder
  Role Play • 15 min
- Practice Quiz: What is RAG?
  Practice Assignment • Grade: 100%

Summary and Evaluation

- **Summary and Highlights: Introduction to RAG**
  Reading • 2 min
- Cheat Sheet: Introduction to RAG
  Reading • 5 min
- Graded Quiz: Introduction to RAG
  Graded Assignment • 30 min

Module 2
Build Apps with RAG

Module 3
Build RAG Apps with LlamaIndex

Summary and Highlights: Introduction to RAG

9 XP • Advanced RAG System Implementation

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Congratulations! You have completed this lesson. At this point in the course, you know that:

- Retrieval-Augmented Generation (RAG) is a machine-learning technique that integrates information retrieval with generative AI to produce accurate and context-aware responses.
- RAG enhances Large Language Models (LLMs) by integrating external or domain-specific knowledge without retraining. This helps LLMs generate more accurate and contextually relevant responses for specialized queries, such as a company's mobile policy.
- RAG consists of two main components: the Retriever, which extracts relevant data from a knowledge base, and the Generator, which uses the retrieved information to generate responses in natural language.
- The RAG process comprises four steps: Text Embedding, Retrieval, Augmented Query Creation, and Model Generation.
- Text Embedding converts user prompts and knowledge base documents into high-dimensional vectors using AI models such as BERT or GPT.
- Retrieval matches the user query with similar vectors from the knowledge base to retrieve relevant information.
- Augmented Query Creation combines retrieved content with the user prompt.
- Model Generation uses the created augmented query to generate a response using the content from the knowledge base.
- Prompt encoding converts a text-based prompt into a numerical representation that an LLM can process. It uses Token Embedding and Vector Averaging to break documents into smaller text chunks, convert them into vectors, and index them in a vector database.
- Distance Metrics measure similarity between user queries and document vectors using methods such as dot product (magnitude-based) or cosine similarity (direction-based).
- RAG is an efficient response generation technique. It retrieves the most relevant text chunks from the knowledge base to augment the model's knowledge and produce an informed response. This ensures responses are accurate, domain-specific, and up-to-date.
- RAG allows chatbots to provide specialized answers by integrating relevant external knowledge, making them more reliable for industry-specific or confidential topics.

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